The anti-supersaturation problem

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Abstract

We introduce a fixed-edge version of inducibility. Given a graph F and an exact n-vertex inducibility maximizer H_n, we study the largest number of induced copies of F in a graph with $|H_n| + q$ edges. We prove a local principle showing that, under perfect stability and a strict one-pair loss condition, every optimizer is obtained from H_n by adding q missing pairs of minimum total loss. For every non-complete Turán graph $F = T(k, r)$, this description is exact throughout a range $1 \le q \le \varepsilon_F n$. We also obtain exact fixed-edge results for unbalanced complete bipartite graphs and for the split graphs $K_{\ell,1,1}$. In each case all equality cases are determined.

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1 Introduction and main result

Mantel's theorem states that every n-vertex graph with more than $\lfloor n^2/4 \rfloor$ edges contains a triangle. Rademacher strengthened this by showing that one additional edge already forces linearly many triangles, and Erdős asked for the corresponding bound when q edges are added above the extremal threshold [5, 6]. Lovász and Simonovits developed this prescribed-edge problem for cliques [14, 15], and Mubayi proved the analogous small-linear-range theorem for every colour-critical graph [17]. These results are typical supersaturation theorems: after the extremal number of edges is exceeded, they minimize the number of copies of a forbidden graph.

We consider the induced analogue. Here adding an edge may destroy induced copies as well as create them. Thus the natural question is different. Once an exact inducibility maximizer is known, how much of the maximum induced-copy count must be lost if the number of edges is prescribed to be slightly larger than that of the maximizer? This is a finite version of the fixed-density questions considered in the feasible-region framework of Liu, Mubayi and Reiher [13].

For graphs F and G, let

$$N(F, G) = \big|\{X \subseteq V(G) : |X| = v(F) \text{ and } G[X] \cong F\}\big|$$

and define

$$I(F, n) = \max\{N(F, G) : v(G) = n\}.$$

We identify a graph with its edge set, write $|G|$ for its number of edges, and write $\overline{G}$ for its set of missing pairs. Suppose that H_n is an exact n-vertex maximizer, so $N(F, H_n) = I(F, n)$, and put $M_n = |H_n|$. For $q \ge 0$, set

$$I(F, n, q) = \max\{N(F, G) : v(G) = n \text{ and } |G| = M_n + q\}.$$

The following finite quantity measures the least possible loss caused by the prescribed extra edges:

$$\Phi(H, F, q) = \min\{N(F, H) - N(F, H + A) : A \subseteq \overline{H}, |A| = q\}. \tag{1.1}$$

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Unlike a first-order expression, $\Phi(H,F,q)$ records all interactions among the added pairs.

We write $T(k,r)$ for the complete r-partite graph on k vertices whose parts are as equal as possible. For $p \ge r$, let

$$\psi_{r,k}(p) = \frac{(p)_r}{p^k}, \qquad (p)_r = p(p-1)\cdots(p-r+1).$$

Liu, Ma and Zhu proved that, whenever $2 \le r < k$, this function has a unique integer maximizer $m = m_{r,k}$ and that $T(n,m)$ is the unique inducibility maximizer for $T(k,r)$ when n is sufficiently large; they also proved the perfect-stability statement used below [12]. Our main result determines the exact optimum at every edge count in a small linear interval above $|T(n,m)|$.

Theorem 1.1. *Let $F = T(k,r)$ with $2 \le r < k$, let $m = m_{r,k}$ be the unique integer maximizing $\psi_{r,k}$, and put $H_n = T(n,m)$ and $M_n = |H_n|$. There is a constant $\varepsilon = \varepsilon(F) > 0$ such that, for all sufficiently large n and every $1 \le q \le \varepsilon n$,*

$$I(F,n,q) = I(F,n) - \Phi(H_n,F,q). \tag{1.2}$$

Moreover, equality is attained precisely by the graphs $H_n + A$ for which $A \subseteq \overline{H_n}$, $|A| = q$, and A attains the minimum in equation (1.1).

Thus no imbalance of the host partition, even when combined with further edge edits, improves on the minimum-loss addition to $T(n,m)$. Expanding the finite loss gives a convenient first-order form. For a pair xy, let $H \oplus xy$ be obtained by toggling xy, and put

$$\Delta(H,F;xy) = N(F,H) - N(F,H \oplus xy), \qquad \lambda_F(n) = \min_{xy \in \overline{H_n}} \Delta(H_n,F;xy).$$

Corollary 1.2. *Under the assumptions of Theorem 1.1, if $q = o(n)$, then*

$$I(F,n,q) = I(F,n) - q\lambda_F(n) + O_F(q^2 n^{k-3}). \tag{1.3}$$

The proof combines two ideas. Perfect stability first places every relevant graph within $O(q)$ edits of a complete multipartite graph. A strict rooted pair calculation then shows that any edit away from this template has loss of order n^{k-2}. The remaining difficulty is that a nearly balanced m-partite graph can have only a quadratic edge deficit. We compare that deficit with the additional loss required to reach the prescribed edge count; this forces the template to be exactly $T(n,m)$. This use of stability follows the symmetrisation framework of Liu, Pikhurko, Sharifzadeh and Staden [11], while the prescribed-edge organization is modeled on the supersaturation argument of Mubayi [17].

For context, inducibility was introduced by Pippenger and Golumbic [19]. Complete bipartite and complete multipartite targets were studied in [3, 4, 2, 8]. Stability methods for induced problems were developed in [18, 11]. Further results on induced cycles and general targets appear in [1, 16, 9, 10, 7, 20].

The paper is organized as follows. In Section 2 we state the fixed-edge principle and two further applications. Its elementary local part is proved in Section 3. Sections 4 and 5 contain the local calculation and the proof of the main theorem. The complete bipartite and split-graph applications are proved in Section 6. The appendix contains the finite cleanup of complete multipartite templates.

2 The fixed-edge principle and further results

2.1 General fixed-edge reduction

We first give a general reduction for fixed-edge inducibility problems. It is stated for a fixed graph F with $f = v(F) \ge 3$; the case $f = 2$ is a constant-scale question about the number of edges and is not the phenomenon considered here. The proof of Theorem 2.3 is given in Section 3, and the proof of the linear control criterion is given in Section 6.2.

Fix a sequence of exact extremal graphs H_n, meaning that $|V(H_n)| = n$ and $N(F,H_n) = I(F,n)$, and put $M_n := |H_n|$. For this fixed sequence, $I(F,n,q)$ denotes the maximum of $N(F,G)$ over all n-vertex graphs with $M_n + q$ edges. For two n-vertex graphs G,H, let $\mathrm{edit}(G,H)$ be the minimum number of edge edits needed to transform G into an isomorphic copy of H. For a family $\mathcal{G}$ of n-vertex graphs, define $\mathrm{edit}(G,\mathcal{G}) :=$ $\min_{H\in\mathcal{G}} \mathrm{edit}(G,H)$. For a graph H and a pair $xy \in \binom{V(H)}{2}$, write $H \oplus xy$ for the graph obtained from H by adding xy if it is missing and deleting xy if it is present, and let

$$\Delta(H,F;xy) := N(F,H) - N(F,H \oplus xy).$$

We use the loss $\Phi(H, F, q)$ defined in equation (1.1).

The first hypothesis is the finite pair-flip part of the strictness condition used in the symmetrisation framework of Liu, Pikhurko, Sharifzadeh and Staden [11].

Definition 2.1. *The sequence (H_n) is* pair-flip strict *if there is $c_{\text{flip}} > 0$ such that, for every sufficiently large n and every pair $xy \in \binom{V(H_n)}{2}$, we have $\Delta(H_n, F; xy) \geq c_{\text{flip}} n^{f-2}$.*

The second hypothesis is a stability input at the prescribed number of edges. It says that any graph whose induced-F count is within the natural first-order error must already be close to the exact extremal graph.

Definition 2.2. *Let $q = q(n)$ satisfy $1 \leq q = o(n)$. The* sublinear closeness condition *holds along q if H_n has at least q missing pairs for all sufficiently large n and, for every $C_0 > 0$, every sequence of n-vertex graphs G_n satisfying*

$$|G_n| = M_n + q \quad \text{and} \quad N(F, G_n) \geq I(F, n) - C_0 q n^{f-2}$$

also satisfies $\operatorname{edit}(G_n, H_n) = o(n)$.

The linear closeness condition *holds above M_n if, for every $C_0 > 0$, there are constants $L_{C_0}, \varepsilon_{C_0} > 0$ such that, for all sufficiently large n and all integers $1 \leq q \leq \varepsilon_{C_0} n$, the graph H_n has at least q missing pairs and every graph G satisfying the two displayed inequalities has* $\operatorname{edit}(G, H_n) \leq L_{C_0} q$.

With these two inputs, the fixed-edge problem is reduced to the finite minimum-loss addition problem over H_n.

Theorem 2.3. *Let F be fixed with $f = v(F) \geq 3$, and let H_n be a sequence of exact extremal graphs with $M_n \coloneqq |H_n|$. Suppose that (H_n) is pair-flip strict. If $q = q(n)$ satisfies $1 \leq q = o(n)$ and the sublinear closeness condition holds along q, then*

$$I(F, n, q) = I(F, n) - \Phi(H_n, F, q) \tag{2.1}$$

for all sufficiently large n. Moreover every extremal graph is isomorphic to $H_n + A$, where A is a q-set of missing pairs of H_n attaining $\Phi(H_n, F, q)$. If the linear closeness condition holds above M_n, then the same conclusion holds throughout a small linear range $1 \leq q \leq \varepsilon n$ for some $\varepsilon > 0$.

The linear closeness condition in Theorem 2.3 is often verified from a stable complete-partite family as follows. Let $\mathcal{B} = (\mathcal{B}_n)$ be a complete-partite family coming from a perfect-stability theorem. A *local parameter system* consists of integer parameter sets Θ_n, graphs $R_n(\theta) \in \mathcal{B}_n$ for $\theta \in \Theta_n$, and distinguished parameter sets $\Theta_n^* \subseteq \Theta_n$ such that $R_n(\theta) \cong H_n$ for every $\theta \in \Theta_n^*$. Write $d_n(\theta) \coloneqq \operatorname{edit}(R_n(\theta), H_n)$.

The system is *linearly controlled by the number of edges* if there are constants $C_E, \eta > 0$ such that, for every sufficiently large n and every $\theta \in \Theta_n$ satisfying $d_n(\theta) \leq \eta n^2$, one has

$$d_n(\theta) \leq C_E \big| |R_n(\theta)| - M_n \big|. \tag{2.2}$$

In the applications below, this inequality follows because every admissible non-trivial local direction changes the number of edges by order n.

We say that this complete-partite family *reduces locally* to the parameter system if, for every pair of constants $C_0, K_0 > 0$, there are constants $C_R, \varepsilon_R > 0$ such that the following holds for all sufficiently large n. Whenever $1 \leq q \leq \varepsilon_R n$ and $B \in \mathcal{B}_n$ satisfies $N(F, B) \geq I(F, n) - C_0 q n^{f-2}$ and $\big| |B| - (M_n + q) \big| \leq K_0 q$, there is $\theta \in \Theta_n$ such that $d_n(\theta) \leq \eta n^2$, $\operatorname{edit}(B, R_n(\theta)) \leq C_R q$, and $\big| |B| - |R_n(\theta)| \big| \leq C_R q$. The extra local condition $d_n(\theta) \leq \eta n^2$ ensures that equation (2.2) is used only where this part-size description is valid. The next theorem packages these two local properties into the linear closeness condition needed by Theorem 2.3.

Theorem 2.4. *Assume perfect stability to $\mathcal{B}$, namely there is C_{st} such that every n-vertex graph G satisfies*

$$\operatorname{edit}(G, \mathcal{B}_n) \leq C_{\text{st}} \frac{I(F, n) - N(F, G)}{n^{f-2}}.$$

Assume that $\mathcal{B}$ reduces locally to a parameter system which is linearly controlled by the number of edges. Then the linear closeness condition above M_n holds in the sense of Definition 2.2.

2.2 Applications beyond the Turán theorem

The same reduction also gives exact fixed-edge results beyond Turán targets. The first case uses a linear edge-count calculation: every non-trivial local change in the part sizes changes the number of edges to first order, so fixing the number of edges forces the correct finite graph.

Unbalanced complete bipartite graphs. Let $1 \le s < t$, put $k := s + t$, and assume

$$s < \binom{t-s}{2}. \tag{2.3}$$

Let $J^{(n)}_{s,t}(x) := \binom{x}{s}\binom{n-x}{t} + \binom{x}{t}\binom{n-x}{s}$ for $0 \le x \le n$. For every sufficiently large n, choose an integer $a_n \le b_n$, $a_n + b_n = n$, such that K_{a_n,b_n} is an exact n-vertex maximizer for $K_{s,t}$. Put $H^{s,t}_n := K_{a_n,b_n}$ and $M^{s,t}_n := |H^{s,t}_n| = a_n b_n$. By Proposition 6.10, the sequence is *linearly unbalanced*: there is $\gamma = \gamma(s,t) > 0$ such that $b_n - a_n \ge \gamma n$ for all sufficiently large n.

The fixed-edge theorem says that no other near-bipartite construction beats the loss $\Phi(H^{s,t}_n, K_{s,t}, q)$ from the best choice of q internal pairs.

Theorem 2.5. *Assume that equation (2.3) holds. There is $\varepsilon = \varepsilon(s,t) > 0$ such that, for all sufficiently large n and all $1 \le q \le \varepsilon n$,*

$$\max\left\{N(K_{s,t}, G) \colon |V(G)| = n \text{ and } |G| = M^{s,t}_n + q\right\} = N(K_{s,t}, H^{s,t}_n) - \Phi(H^{s,t}_n, K_{s,t}, q).$$

Every extremal graph is isomorphic to a graph obtained from $H^{s,t}_n$ by adding a q-set of missing internal pairs attaining $\Phi(H^{s,t}_n, K_{s,t}, q)$.

Split graphs $K_{\ell,1,1}$. The next case uses a quadratic splitting step: secondary non-singleton parts can be split into singleton parts at controlled edit cost while increasing the number of copies by a quadratic amount. For integers $a + b = n$, let $S_{a,b}$ be the *split graph* obtained as the join of an independent set A of size a and a clique B of size b. Then $N(K_{\ell,1,1}, S_{a,b}) = \binom{a}{\ell}\binom{b}{2}$. Let a_n maximize $\binom{a}{\ell}\binom{n-a}{2}$, set $b_n := n - a_n$, $H^\ell_n := S_{a_n,b_n}$, and $M^\ell_n := |H^\ell_n| = \binom{n}{2} - \binom{a_n}{2}$. By Proposition 6.16, these are precisely the finite exact extremal graphs for all sufficiently large n, and $a_n/n \to \ell/(\ell+2)$ and $b_n/n \to 2/(\ell+2)$.

The missing pairs of H^ℓ_n are exactly the pairs inside its independent part A. Thus the relevant loss is $\Phi(H^\ell_n, K_{\ell,1,1}, q)$. The proof first uses a quadratic reduction to reach the split family.

Theorem 2.6. *For every fixed $\ell \ge 3$, there is $\varepsilon = \varepsilon(\ell) > 0$ such that, for all sufficiently large n and all $1 \le q \le \varepsilon n$,*

$$\max\left\{N(K_{\ell,1,1}, G) \colon |V(G)| = n \text{ and } |G| = M^\ell_n + q\right\} = N(K_{\ell,1,1}, H^\ell_n) - \Phi(H^\ell_n, K_{\ell,1,1}, q).$$

Every extremal graph is isomorphic to a graph obtained from H^ℓ_n by adding a q-set of pairs inside the independent part A attaining $\Phi(H^\ell_n, K_{\ell,1,1}, q)$. If $q = o(n)$, then

$$\Phi(H^\ell_n, K_{\ell,1,1}, q) = q\binom{a_n - 2}{\ell - 2}\binom{b_n}{2} + O_\ell(q^2 n^{\ell-1}).$$

3 The local fixed-edge argument

Throughout this section, F is fixed with $f = v(F) \ge 3$, and H_n and M_n are as in Section 2. We prove the general reduction from Section 2. The first two elementary estimates control how one-edge losses change under a small number of preliminary edits.

Lemma 3.1. *There is a constant C_F depending only on F such that the following holds. Let H, J be two n-vertex graphs with $|H \triangle J| \le t$. If a pair xy has the same adjacency status in H and J, then $|\Delta(H, F; xy) - \Delta(J, F; xy)| \le C_F t n^{f-3}$.*

Proof. The two losses differ only on f-vertex sets which contain xy and also contain at least one pair from $H \triangle J$. Fix such a modified pair uv. Since xy itself is not modified, the set $\{x, y, u, v\}$ has at least three vertices. Hence the number of f-sets containing both xy and uv is at most a constant depending only on F times n^{f-3}. Summing over the at most t modified pairs proves the estimate. Overcounting is harmless because only an upper bound is needed. □

The next estimate sums these marginal losses over a set of additions. It is the place where the finite loss $\Phi(H, F, q)$ becomes comparable to a first-order sum.

Lemma 3.2. *Let H be an n-vertex graph and let A be a set of q missing pairs of H. Then*

$$N(F,H) - N(F,H+A) = \sum_{xy\in A} \Delta(H,F;xy) + O_F(q^2 n^{f-3}). \tag{3.1}$$

In particular, if $q = o(n)$ and all one-edge losses under consideration are $O_F(n^{f-2})$, then the error is $o(qn^{f-2})$.

Proof. Order A as $e_1, \dots, e_q$, and let $H_i \coloneqq H + e_1 + \cdots + e_i$. Before adding e_i, the current graph H_{i-1} differs from H in $i-1$ pairs, while e_i is still missing in both graphs. Lemma 3.1 gives $\Delta(H_{i-1}, F; e_i) = \Delta(H, F; e_i) + O_F(in^{f-3})$. Summing over i gives equation (3.1). ☐

We now use pair-flip strictness to remove all deleted edges from a near copy of H_n. The point is that any detour which deletes an edge of H_n and compensates by adding another missing pair is strictly worse than keeping only the required q additions.

Proposition 3.3. *Assume pair-flip strictness with constant c_{flip}. Let $q \coloneqq q_n$ and $\ell_n = o(n)$. Suppose $\mathrm{edit}(G, H_n) \le \ell_n$ and $|G| = M_n + q$. Then, for all sufficiently large n, $N(F,G) \le I(F,n) - \Phi(H_n, F, q)$. Equality holds if and only if $G \cong H_n + A$ for a q-set A of missing pairs of H_n attaining $\Phi(H_n, F, q)$.*

Proof. Put $H \coloneqq H_n$. After relabeling G, write $G = H + A - D$, where $A \subseteq \overline{H}$, $D \subseteq H$, and $A \cap D = \varnothing$. Then $|A| + |D| \le \ell_n$ and $|A| - |D| = q$. If $D = \varnothing$, then $|A| = q$, and the definition of $\Phi(H, F, q)$ gives $N(F,G) = N(F, H+A) \le I(F,n) - \Phi(H,F,q)$, with equality precisely when A attains $\Phi(H,F,q)$.

Assume $D \ne \varnothing$ and put $r_0 \coloneqq |D|$, so $|A| = q + r_0$. Choose a q-subset $A_0 \subseteq A$ and put $A_1 \coloneqq A \setminus A_0$. Let $J_0 \coloneqq H + A_0$. Obtain G from J_0 by adding all pairs of A_1 and deleting all pairs of D, in any order. At every intermediate graph J we have $|J \triangle H| \le |A| + |D| \le \ell_n$, and immediately before each flip the flipped pair has the same adjacency status in J as in H. By Lemma 3.1, $\Delta(J, F; z) \ge c_{\mathrm{flip}} n^{f-2} - C_F \ell_n n^{f-3} > 0$ for all sufficiently large n. Thus every one of the $2r_0$ flips strictly decreases $N(F, \cdot)$. Consequently $N(F,G) < N(F, J_0) = N(F, H + A_0) \le I(F,n) - \Phi(H,F,q)$, proving both the strict inequality when $D \ne \varnothing$ and the equality statement. ☐

The same argument gives a small linear range once the preliminary edit distance is only $O(q)$.

Proposition 3.4. *Assume pair-flip strictness with constant c_{flip}. For every $L > 0$ there is $\varepsilon_L > 0$ such that the conclusion of Proposition 3.3 holds whenever $\mathrm{edit}(G, H_n) \le Lq$, $|G| = M_n + q$, and $1 \le q \le \varepsilon_L n$.*

Proof. The proof is identical to that of Proposition 3.3. The only changed estimate is $\Delta(J, F; z) \ge c_{\mathrm{flip}} n^{f-2} - C_F L q n^{f-3}$, which is positive for all $q \le \varepsilon_L n$ once $\varepsilon_L < c_{\mathrm{flip}}/(2C_F L)$. ☐

Proof of Theorem 2.3. Changing one adjacency affects only $O_F(n^{f-2})$ induced F-copies, so $\Phi(H_n, F, q) = O_F(qn^{f-2})$. Choose a q-set A_0 of missing pairs of H_n attaining $\Phi(H_n, F, q)$. Then $H_n + A_0$ has the target number of edges and gives the lower bound $\max_{|V(G)|=n,\ |G|=M_n+q} N(F,G) \ge I(F,n) - \Phi(H_n, F, q)$. Let G be extremal among graphs with $|G| = M_n + q$. The lower bound implies $N(F,G) \ge I(F,n) - O_F(qn^{f-2})$. The sublinear closeness condition gives $\mathrm{edit}(G, H_n) = o(n)$. Proposition 3.3 gives $N(F,G) \le I(F,n) - \Phi(H_n, F, q)$. This proves the exact value and equality characterisation. Under the linear closeness condition, the same argument gives $\mathrm{edit}(G, H_n) \le Lq$ for $q \le \varepsilon n$, and choosing ε small enough for Proposition 3.4 proves the small linear statement. ☐

4 Local strictness for Turán targets

This section proves the local pair-flip estimate needed for non-complete Turán targets. Write $k = ar + b$ with $a \ge 1$ and $0 \le b < r$. The independent class sizes of $F = T(k, r)$ are $\alpha_1, \dots, \alpha_r \in \{a, a+1\}$, with exactly b of them equal to $a+1$. Let

$$A_F \coloneqq \sum_{i\in[r]} \alpha_i(\alpha_i - 1), \qquad B_F \coloneqq 2 \sum_{\{i,j\}\in\binom{[r]}{2}} \alpha_i\alpha_j = k^2 - \sum_{i\in[r]} \alpha_i^2, \tag{4.1}$$

and let $u_1 \coloneqq |\{i \colon \alpha_i = 1\}|$ and $u_2 \coloneqq |\{i \colon \alpha_i = 2\}|$.

We first prove two elementary consequences of the definition of m. Let $U_r \coloneqq \sum_{i\in[r-1]} i = r(r-1)/2$ and $V_r \coloneqq \sum_{i\in[r-1]} i^2 = r(r-1)(2r-1)/6$.

Lemma 4.1. *Let m be the unique integer maximizing $\psi_{r,k}(p) := (p)_r/p^k$ over $p \ge r$. Then $m - r \le \lceil U_r/(k-r)\rceil$. If $k = r + b$ with $1 \le b \le r-1$, then*

$$m > r - 1 + \frac{(r-b)(r-b-1)}{2b}. \tag{4.2}$$

Proof. Extend $\psi_{r,k}$ to $x > r-1$ and let $\varphi(x) := \log \psi_{r,k}(x) = \sum_{j=0}^{r-1} \log(x-j) - k\log x$. Then $\varphi'(x) = x^{-1}(S_r(x) - (k-r))$, where $S_r(x) := \sum_{i\in[r-1]} i/(x-i)$. Since S_r is strictly decreasing on $x > r-1$, the function $\psi_{r,k}$ is increasing up to the unique root ξ of $S_r(\xi) = k - r$ and decreasing after that root. Hence an integer maximizer lies at distance at most one from ξ, in particular $m \le \lceil \xi \rceil$. As $S_r(x) \le U_r/(x-r+1)$, every $x > r-1+U_r/(k-r)$ satisfies $S_r(x) < k-r$. Hence $\xi \le r-1+U_r/(k-r)$, and therefore $m - r \le \lceil U_r/(k-r)\rceil$.

Now assume $k = r+b$ with $1 \le b \le r-1$. The case $r = 2, b = 1$ is immediate from $m \ge r$, because the right hand side of equation (4.2) is 1. Suppose $r \ge 3$. Let $L := (r-b)(r-b-1)/(2b)$ and set $x := r + L$. By Cauchy's inequality,

$$S_r(x) = \sum_{i\in[r-1]} \frac{i^2}{i(x-i)} \ge \frac{U_r^2}{xU_r - V_r}. \tag{4.3}$$

A direct simplification gives $U_r^2 + bV_r - bxU_r = br(r-1)(4r-5-3b)/12$, which is positive because $b \le r-1$ and $r \ge 3$. Thus $S_r(x) > b$, so the root ξ satisfies $\xi > x = r + L$. If an integer maximizer had $m \le r-1+L$, then $m+1 \le r+L < \xi$, and the strict increase of $\psi_{r,k}$ on $(r-1,\xi)$ would give $\psi_{r,k}(m+1) > \psi_{r,k}(m)$, a contradiction. This proves equation (4.2). □

We next compute the leading loss of the two flips away from a balanced complete m-partite graph. Label the independent classes of F as $C_1, \dots, C_r$, where $|C_i| = \alpha_i$. A *typed copy* is an ordinary induced copy together with a size-preserving identification of its independent classes with $C_1, \dots, C_r$. If μ_d denotes the number of classes of F of size d, then every ordinary copy has exactly $\prod_d \mu_d!$ typings. Thus typed and ordinary losses differ by this fixed positive factor. In a complete multipartite host, counting a typed copy amounts to assigning its labeled classes to the relevant host parts, except for the two classes incident with the flipped pair.

Lemma 4.2. *Let $B := K_{N,\dots,N}$ be the balanced complete m-partite graph with labeled host parts. Put $W_F := \prod_{i\in[r]}(\alpha_i!)^{-1}$. If xy is a missing pair inside one host part, then adding xy changes the typed induced-F count by*

$$W_F(m-1)_{r-2}\big((m-r+1)A_F - u_1(u_1-1)\big)N^{k-2} + O_F(N^{k-3}) \tag{4.4}$$

in the loss direction. If xy is a present pair between two distinct host parts, then deleting xy changes the typed count by

$$W_F(m-2)_{r-2}\big(B_F - 2u_2(m-r)\big)N^{k-2} + O_F(N^{k-3}) \tag{4.5}$$

in the loss direction.

Proof. For an internal pair xy, the destroyed typed copies are exactly those in which x, y lie in the same independent class of F. If the class is i, there are $\binom{N-2}{\alpha_i-2}$ choices for the remaining vertices of that class and, to leading order, $(m-1)_{r-1}\prod_{j\in[r]\setminus\{i\}}\binom{N}{\alpha_j}$ choices for the other classes. Summing over i gives leading destroyed coefficient $W_F(m-1)_{r-1}A_F$. The only typed copies created by adding the internal edge are those in which x and y represent two singleton classes of F: if either class had another vertex, that vertex would have the wrong adjacency to one of x, y. Hence the leading created coefficient is $W_F(m-1)_{r-2}u_1(u_1-1)$. Subtracting gives equation (4.4).

For a cross pair xy, the destroyed typed copies are exactly those in which x and y lie in two different independent classes. Summing over ordered choices of these classes gives leading coefficient $W_F(m-2)_{r-2}B_F$. The only typed copies created by deleting the cross edge are those in which x, y become the two vertices of a class of size two; no class of size at least three can be created, because a third vertex in the same class would be adjacent to one of x, y. For each size-two class the factor relative to W_F is 2, and the other $r-1$ classes may be assigned to $(m-2)_{r-1}$ host parts. Thus the leading created coefficient is $2W_Fu_2(m-2)_{r-1}$. Subtracting gives equation (4.5). □

It remains to show that the two leading coefficients just computed are positive for the actual maximizing value of m. This is the only place where the elementary bounds on m from Lemma 4.1 are needed.

Lemma 4.3. *For every $2 \le r < k$ and every maximizing integer m as above,*

$$(m-r+1)A_F - u_1(u_1-1) > 0, \qquad B_F - 2u_2(m-r) > 0. \tag{4.6}$$

Proof. We first prove the internal inequality. If $a \ge 2$, then $u_1 = 0$ and $A_F > 0$. If $a = 1$, then $k = r + b$ with $1 \le b \le r-1$, $A_F = 2b$, and $u_1 = r - b$. The internal inequality becomes $2b(m-r+1) > (r-b)(r-b-1)$, which is exactly Lemma 4.1.

We next prove the cross inequality. If $a \ge 3$, then $u_2 = 0$ and $B_F > 0$. If $a = 2$, then $u_2 = r - b$ and $B_F = (2r+b)^2 - 4(r-b) - 9b$. A direct calculation gives $B_F - 4(r-1)(r-b) = b(b+8r-9) \ge 0$, so $B_F/(2u_2) \ge 2(r-1)$ when $u_2 > 0$. Lemma 4.1 gives $m - r \le \lceil U_r/(r+b) \rceil \le r-1$, and hence $m - r < 2(r-1) \le B_F/(2u_2)$ for $r \ge 2$. Finally suppose $a = 1$. Then $u_2 = b$ and $B_F = (r+b)^2 - (r-b) - 4b$, so

$$\frac{B_F}{2b} = \frac{U_r}{b} + r + \frac{b-3}{2}. \tag{4.7}$$

The last two terms on the right have sum at least 1 for $r \ge 2$ and $1 \le b \le r-1$, while Lemma 4.1 gives $m - r \le \lceil U_r/b \rceil$. Since the right hand side of equation (4.7) is strictly larger than $\lceil U_r/b \rceil$, we get $B_F > 2b(m-r) = 2u_2(m-r)$. □

The coefficient calculation gives strict losses in the perfectly balanced graph. The next proposition is the robust form used later: the losses survive small part-size imbalance and a small number of preliminary edits.

Proposition 4.4. *Let $F \coloneqq T(k,r)$ with $2 \le r < k$, and let $m \coloneqq m(F)$ be the unique integer maximizing $\psi_{r,k}(p)$ over $p \ge r$. There are constants $c_F, \eta_F > 0$ such that the following holds for all sufficiently large n. Let $B \coloneqq K_{b_1,\dots,b_m}$ be a complete m-partite graph on n vertices with $|b_i - n/m| \le \eta_F n$ for every i. Let J be obtained from B by at most $\eta_F n$ edge edits. If xy has the same adjacency status in J as in B, then every flip of xy away from B has loss at least $c_F n^{k-2}$. Equivalently, adding an unedited missing pair inside a part of B, or deleting an unedited cross-pair of B, decreases the number of induced copies of F by at least $c_F n^{k-2}$.*

Proof. For the perfectly balanced part vector, Lemmas 4.2 and 4.3 give a positive leading coefficient for both types of flips away from B. The same conclusion holds for the ordinary untyped count. Indeed, if the multiset of part sizes of F has multiplicities $\mu_1, \dots, \mu_s$, then every ordinary induced copy corresponds to $\prod_{j\in[s]} \mu_j!$ typed copies. Passing from typed to ordinary counts therefore only divides every rooted loss by this fixed positive integer. For part sizes satisfying $|b_i - n/m| \le \eta n$, the rooted loss is a polynomial in the variables b_i of degree at most $k-2$, plus an $O_F(n^{k-3})$ rounding error. By continuity of the leading homogeneous polynomial and compactness of the part-size simplex near $(1/m, \dots, 1/m)$, there are $c_0 > 0$ and $\eta_0 > 0$ such that every away flip in B has loss at least $2c_0 n^{k-2}$ whenever $\eta \le \eta_0$ and n is large.

Now let J differ from B in at most ηn pairs, and let xy be a pair whose status in J agrees with its status in B. Lemma 3.1, applied with $f = k$, gives $|\Delta(J,F;xy) - \Delta(B,F;xy)| \le C_F \eta n^{k-2}$. Choose $\eta_F \le \eta_0$ so small that $C_F \eta_F \le c_0$, and set $c_F \coloneqq c_0$. Then the loss of the away flip in J is at least $c_F n^{k-2}$. □

We record the direction convention explicitly, since the later fixed-edge argument only flips pairs away from the underlying complete multipartite graph.

Remark 4.5. *The direction of the flip is essential. If a preliminary edit has already added an internal pair, then deleting that pair moves back toward the complete multipartite graph and may increase the number of copies. Proposition 4.4 is exactly the form needed in the proof with a fixed number of edges: every flip used there changes a pair whose current status still agrees with the underlying complete multipartite graph, and flips it away from that graph.*

5 The balanced comparison and proof of the main theorem

We use the following form of the exact and stability theorem of Liu–Ma–Zhu [12, Theorems 1.1 and 1.5]. The quoted input includes the uniqueness of the integer maximizer $m \coloneqq m(F)$ in the Turán-inducibility problem; this is what allows the proof below to work with the single exact graph $T(n,m)$ rather than several possible part-size types.

Theorem 5.1 (Liu–Ma–Zhu)**.** *Let $F \coloneqq T(k,r)$ with $2 \le r < k$, and let $m \coloneqq m(F)$ be the unique integer maximizing $\psi_{r,k}(p) \coloneqq (p)_r/p^k$ over $p \ge r$. There are constants C_{T} and n_{T}, depending only on F, such that the following holds for every $n \ge n_{\mathrm{T}}$.*

First, $T(n,m)$ is the unique n-vertex maximizer of the function $G \mapsto N(F,G)$. Second, for every n-vertex graph G, there is a complete multipartite n-vertex graph C such that

$$\mathrm{edit}(G,C) \le C_{\mathrm{T}} \frac{I(F,n) - N(F,G)}{n^{k-2}}.$$

We first convert the quoted stability theorem into the finite reduction needed in the balanced comparison. The output is not yet closeness to $T(n,m)$, but closeness to some balanced complete m-partite graph.

Lemma 5.2. *Let $F := T(k,r)$, let $m := m(F)$, let $H_n := T(n,m)$, and put $M_n := |H_n|$. For every $C_0 > 0$ and every $\zeta > 0$, there are constants C_R, ε_R, and n_R, depending only on F, C_0, ζ, such that the following holds for all $n \ge n_R$.*

If $1 \le q \le \varepsilon_R n$, $|G| = M_n + q$, and $N(F,G) \ge I(F,n) - C_0 q n^{k-2}$, then G is within $C_R q$ edge edits of a complete m-partite graph $B = K_{b_1,\dots,b_m}$ satisfying $|b_i - n/m| \le \zeta n$ for every i.

Proof. By Theorem 5.1, there is a complete multipartite graph C such that $\mathrm{edit}(G,C) \le C_1 q$. Since one edge edit changes $N(F,\cdot)$ by at most $C_F n^{k-2}$, we have

$$N(F,C) \ge I(F,n) - C_2 q n^{k-2}.$$

Apply Proposition A.3 from Appendix A, with $D = C_2 q$. After decreasing ε_R and increasing n_R, it gives a complete m-partite graph $B = K_{b_1,\dots,b_m}$ such that $\mathrm{edit}(C,B) \le C_3 q$ and $|b_i - n/m| \le \zeta n$ for every i. The triangle inequality gives $\mathrm{edit}(G,B) \le (C_1 + C_3) q$. $\square$

Let $H_n := T(n,m)$ have part sizes $h_1, \dots, h_m$, and put $M_n := |H_n|$. For a complete m-partite graph $B := K_{b_1,\dots,b_m}$ on n vertices, define the *balanced edge deficit*

$$L(B) := M_n - |B|.$$

Equivalently,

$$L(B) = \frac{1}{2} \left(\sum_{i \in [m]} b_i^2 - \sum_{i \in [m]} h_i^2 \right).$$

Thus $L(B) \ge 0$, and $L(B) = 0$ holds if and only if $B \cong H_n$.

The first elementary estimate turns this edge deficit into part-size control.

Lemma 5.3. *After relabeling the parts of B,*

$$\sum_{i \in [m]} |b_i - h_i| \le C_m \sqrt{L(B) + 1}.$$

Proof. Write $b_i := n/m + x_i$ and $h_i := n/m + \theta_i$. Then $\sum_{i \in [m]} x_i = \sum_{i \in [m]} \theta_i = 0$, and each θ_i is bounded in absolute value by 1. Hence

$$2L(B) = \sum_{i \in [m]} x_i^2 - \sum_{i \in [m]} \theta_i^2.$$

It follows that $\sum_{i \in [m]} x_i^2 \le 2L(B) + O_m(1)$. Since $b_i - h_i = x_i - \theta_i$, we get $\sum_{i \in [m]} (b_i - h_i)^2 \le C_m (L(B) + 1)$. Cauchy's inequality gives the claim. $\square$

The next estimate says that the finite loss $\Phi(\cdot, F, q)$ changes only mildly when two complete m-partite graphs have almost the same part sizes.

Lemma 5.4. *Let F be a graph with $f = v(F)$, and fix $m \ge 2$ and $\alpha > 0$. There are constants C and $\varepsilon_0 > 0$, depending only on F, m, α, such that the following holds.*

Let $B := K_{b_1,\dots,b_m}$ and $H := K_{h_1,\dots,h_m}$ be complete m-partite graphs on the same n-vertex set, with $b_i, h_i \ge \alpha n$ for every i. After relabeling the parts, put $S := \sum_{i \in [m]} |b_i - h_i|$. If $q \le \varepsilon_0 n$, then

$$|\Phi(B,F,q) - \Phi(H,F,q)| \le C q S n^{f-3}.$$

Proof. We prove one inequality; the other follows by interchanging B and H. Since $\Phi(\cdot, F, q)$ is invariant under relabeling, we may realize the two multipartite graphs on the same vertex set in such a way that, after relabeling the parts, $|B_i \cap H_i| = \min\{b_i, h_i\}$ for every i. Put $C_i := B_i \cap H_i$ and $W := V \setminus \bigcup_{i\in[m]} C_i$. Then $|W| = \frac{1}{2}\sum_{i\in[m]} |b_i - h_i| \le S$. Let A be a q-set of missing pairs of H attaining $\Phi(H, F, q)$. In the i-th part, the abstract pair pattern of A uses at most $2q$ vertices. Choose ε_0 so small that $2\varepsilon_0 < \alpha/2$. Since $|C_i| \ge \alpha n$ for every i and $q \le \varepsilon_0 n$, the same abstract pair pattern can be realized inside the cores C_i. By symmetry inside the parts of H, this realization has the same loss as A; call it again A. Now view the same set of pairs as a q-set of missing pairs in B, and call it A'.

The graphs H and B agree on all pairs whose endpoints both lie in $\bigcup_{i\in[m]} C_i$. Hence the two losses $N(F, H) - N(F, H + A)$ and $N(F, B) - N(F, B + A')$ can differ only on f-sets which contain at least one added pair and at least one vertex of W. For each added pair and each choice of a vertex of W, there are $O_F(n^{f-3})$ choices for the remaining vertices. Therefore the two losses differ by at most $C_F q|W|n^{f-3} \le C_F qSn^{f-3}$. It follows that $\Phi(B, F, q) \le N(F, B) - N(F, B + A') \le \Phi(H, F, q) + C_F qSn^{f-3}$. Interchanging B and H gives the reverse inequality. $\square$

We now prove the main theorem. The construction $H_n + A$ gives the required lower bound. The heart of the proof is to show that the complete-partite template supplied by stability is exactly balanced; the prescribed extra edges then cannot be offset by deletions elsewhere.

Proof of Theorem 1.1. Fix $F = T(k, r)$, put $f = k$, let $m = m_{r,k}$, and write $H_n = T(n, m)$ and $M_n = |H_n|$. The exact extremal and stability input is Theorem 5.1, and the robust pair-flip estimate is Proposition 4.4. Choose $\zeta > 0$ small enough for that estimate and for $\zeta < 1/(3m)$. Apply Lemma 5.2 with this value of ζ, and let C_R be the resulting constant. We choose $\varepsilon > 0$ sufficiently small in terms of F, ζ, C_R and all constants in Lemma 5.4 and Proposition 4.4; finally, n is chosen sufficiently large. These choices are independent of q, so all estimates below hold uniformly for $1 \le q \le \varepsilon n$.

Let G be extremal among graphs with $|G| = M_n + q$. The construction $H_n + A$, where A attains $\Phi(H_n, F, q)$, gives

$$N(F, G) \ge I(F, n) - \Phi(H_n, F, q) \ge I(F, n) - C_0 qn^{f-2}$$

for a constant $C_0 = C_0(F)$. After decreasing ε if necessary, H_n has at least q missing pairs for every $q \le \varepsilon n$ and all sufficiently large n, so this construction and the quantity $\Phi(H_n, F, q)$ are well defined.

The finite reduction gives a complete m-partite graph $B = K_{b_1,\dots,b_m}$ such that $\mathrm{edit}(G, B) \le C_R q$ and $|b_i - n/m| \le \zeta n$ for every i. Since $|G| - |B| = q + L(B)$, we have $q + L(B) \le C_R q$. In particular, $L(B) = O(q)$. Our choice of ε ensures that $q + L(B)$ is smaller than the number of missing pairs in B, so $\Phi(B, F, q + L(B))$ is well defined.

After relabeling, write $G = B + A - D$, where A consists of missing pairs of B and D consists of present pairs of B. Then $|A| - |D| = q + L(B)$ and $|A| + |D| \le C_R q$.

We first compare G with a pure addition over B. If $D = \varnothing$, then $|A| = q + L(B)$, and G is such a pure addition. If $D \ne \varnothing$, choose $A_0 \subseteq A$ with $|A_0| = q + L(B)$. Starting from $B + A_0$, obtain G by adding all pairs in $A \setminus A_0$ and deleting all pairs in D. Every intermediate graph is within $C_R q$ edits of B. Since $C_R q \le \zeta n$, each added pair from $A \setminus A_0$ is still a missing pair of both the intermediate graph and B before it is added, and each deleted pair from D is still a present pair of both the intermediate graph and B before it is deleted. Thus every such flip is an unedited away-from-B flip and decreases $N(F, \cdot)$. Therefore

$$N(F, G) \le N(F, B) - \Phi(B, F, q + L(B)) \le I(F, n) - \Phi(B, F, q + L(B)).$$

We compare the finite loss over B with the finite loss over the balanced graph H_n. Let $S := \sum_{i\in[m]} |b_i - h_i|$. Lemma 5.3 gives $S \le C_m\sqrt{L(B) + 1}$. Since $\zeta < 1/(3m)$ and n is large, all parts of both B and H_n have size at least αn, where $\alpha = 1/(3m)$. Hence Lemma 5.4 gives

$$\Phi(B, F, q) \ge \Phi(H_n, F, q) - Cq\sqrt{L(B) + 1}\, n^{f-3}.$$

The loss is also monotone, with a quantitative gap, as the number of added pairs increases. Let A^* be a $(q + L(B))$-set attaining $\Phi(B, F, q + L(B))$, and choose any q-subset $A_1 \subseteq A^*$. Add the remaining $L(B)$ pairs one at a time. Before each of these additions, the current graph differs from B in at most $q + L(B) \le C_R q$ pairs, and the pair currently being added is still missing in both the current graph and B. With ε chosen smaller if necessary, the local strictness estimate gives a marginal loss at least $(c/2)n^{f-2}$ for each of these $L(B)$ unedited additions away from B. Since the loss of A_1 is at least $\Phi(B,F,q)$, we have

$$\Phi(B,F,q+L(B)) \geq \Phi(B,F,q) + \frac{c}{2}L(B)n^{f-2}.$$

Combining the last two estimates gives

$$\Phi(B,F,q+L(B)) \geq \Phi(H_n,F,q) + \frac{c}{2}L(B)n^{f-2} - Cq\sqrt{L(B)+1}\,n^{f-3}.$$

If $L(B) \geq 1$, the error term divided by $L(B)n^{f-2}$ is at most $Cq/(n\sqrt{L(B)})$, which is at most $C\varepsilon$. Choosing ε small enough gives

$$\Phi(B,F,q+L(B)) > \Phi(H_n,F,q).$$

This contradicts the lower-bound construction. Hence $L(B)=0$.

When $L(B)=0$, the balanced edge-deficit identity gives $B \cong H_n$. Thus $\mathrm{edit}(G,H_n) \leq C_R q$. Proposition 3.4 now applies, after decreasing ε once more. It gives

$$N(F,G) \leq I(F,n) - \Phi(H_n,F,q),$$

with equality only when $G \cong H_n + A$ for a q-set A attaining $\Phi(H_n,F,q)$. This proves both equation (1.2) and the equality statement. □

Proof of Corollary 1.2. Let $\lambda_F(n) := \min\left\{\Delta(H_n,F;xy)\colon xy \in \overline{H}_n\right\}$.

For the lower bound on $\Phi(H_n,F,q)$, let A be any q-set of missing pairs of H_n. Lemma 3.2 gives

$$N(F,H_n) - N(F,H_n+A) = \sum_{xy\in A} \Delta(H_n,F;xy) + O_F(q^2n^{k-3}).$$

Every summand is at least $\lambda_F(n)$, so

$$N(F,H_n) - N(F,H_n+A) \geq q\lambda_F(n) - O_F(q^2n^{k-3}).$$

Taking the minimum over A gives the same lower bound for $\Phi(H_n,F,q)$.

For the upper bound, choose a missing pair xy of H_n with $\Delta(H_n,F;xy) = \lambda_F(n)$. This pair lies inside one part of H_n, and that part contains $\Omega(n^2)$ missing pairs with the same one-edge loss. Since $q=o(n)$, choose q distinct missing pairs of this same type. Lemma 3.2 gives

$$\Phi(H_n,F,q) \leq q\lambda_F(n) + O_F(q^2n^{k-3}).$$

Thus

$$\Phi(H_n,F,q) = q\lambda_F(n) + O_F(q^2n^{k-3}).$$

Substituting this into Theorem 1.1 proves equation (1.3). □

6 Complete-partite reductions and further applications

The all-Turán theorem uses the balanced part-size comparison, but the same fixed-edge argument applies more broadly. This section proves the linear control criterion from Section 2, gives the quadratic splitting argument, and proves the finite estimates needed for the two applications. Throughout this section, fix an exact extremal sequence H_n and put $M_n := |H_n|$.

6.1 The complete-partite limit space

We record the compactness fact used in both applications. Let

$$\mathcal{P} = \left\{ x=(x_1,x_2,\ldots) : x_1 \geq x_2 \geq \cdots \geq 0 \text{ and } \sum_{i\geq 1} x_i \leq 1 \right\}, \qquad x_0 = 1 - \sum_{i\geq 1} x_i.$$

Here the positive coordinates represent non-singleton parts and x_0 represents singleton mass. We equip $\mathcal{P}$ with the complete-partite edit metric of Liu, Pikhurko, Sharifzadeh and Staden [11, Sections 2.1–2.2]. In this metric $\mathcal{P}$ is compact, and every induced-density polynomial of a fixed complete multipartite graph is continuous [11, Lemma 2.5 and Corollary 3.2]. For a finite complete multipartite graph C, its partite vector $x(C)$ is obtained by dividing the sizes of its non-singleton parts by $v(C)$ and listing them in non-increasing order.

Lemma 6.1 (Finite localization)**.** *Let F be a fixed complete multipartite graph, and suppose that its complete-partite limit problem has the unique maximizer $x^* \in \mathcal{P}$. For every neighborhood U of x^* there are $\delta > 0$ and n_0 such that, whenever $n \geq n_0$ and C is an n-vertex complete multipartite graph satisfying*

$$N(F, C) \geq I(F, n) - \delta n^{v(F)},$$

we have $x(C) \in U$.

Proof. Otherwise there is a sequence C_j with $v(C_j) \to \infty$, with partite vectors outside U, and with $I(F, v(C_j)) - N(F, C_j) = o(v(C_j)^{v(F)})$. Compactness gives a convergent subsequence $x(C_j) \to y \in \mathcal{P} \setminus U$. The finite copy count, after division by $v(C_j)^{v(F)}$, differs from its limit polynomial by $O_F(1/v(C_j))$. Brown–Sidorenko symmetrisation [4, Proposition 1] identifies the unrestricted limiting maximum with the complete-partite maximum. Continuity therefore makes y a limiting maximizer, contrary to uniqueness. □

We also use the following stability criterion. We state it in the form needed here and retain the terminology of the cited paper. A pair flip has scale n^{f-2}, whereas changing the attachment of one vertex in d incident pairs has scale dn^{f-2}.

Theorem 6.2 (Liu–Pikhurko–Sharifzadeh–Staden)**.** *Let F be a fixed complete multipartite graph with $f = v(F)$. Suppose that the induced-F density is symmetrisable, that its set of complete-partite limit maximizers is finite, and that every maximizer satisfies conditions* (Str1) *and* (Str2) *of [11, Definition 5]. Namely, every non-trivial pair flip has a strictly negative limiting derivative, and every one-vertex extension that is not a clone has a rooted loss bounded below by a positive constant times its normalized incident edit distance from the clone set. Then there is a constant $C = C(F)$ such that every n-vertex graph G is within*

$$C \, \frac{I(F, n) - N(F, G)}{n^{f-2}}$$

edge edits of a complete multipartite graph.

This is the unnormalized form of [11, Theorem 1.1]. The two scales in the statement explain the estimates verified in Lemmas 6.14 and 6.15 below.

6.2 Linear edge control

We first prove Theorem 2.4.

Proof of Theorem 2.4. Let G satisfy $|G| = M_n + q$ and $N(F, G) \geq I(F, n) - C_0 q n^{f-2}$, where $1 \leq q \leq \varepsilon n$ and ε will be chosen small. Perfect stability gives $B \in \mathcal{B}_n$ with $\operatorname{edit}(G, B) = O(q)$. Changing $O(q)$ adjacencies changes $N(F, \cdot)$ by $O(qn^{f-2})$, and hence

$$N(F, B) \geq I(F, n) - O(qn^{f-2}) \quad \text{and} \quad \big|\, |B| - (M_n + q) \,\big| = O(q).$$

The local reduction gives $\theta \in \Theta_n$ such that $\operatorname{edit}(B, R_n(\theta)) = O(q)$, $\big|\, |B| - |R_n(\theta)| \,\big| = O(q)$, and $d_n(\theta) \leq \eta n^2$. Therefore

$$\big|\, |R_n(\theta)| - M_n \,\big| \leq \big|\, |R_n(\theta)| - |B| \,\big| + \big|\, |B| - |G| \,\big| + q = O(q).$$

By equation (2.2), $\operatorname{edit}(R_n(\theta), H_n) = d_n(\theta) \leq C_E \big|\, |R_n(\theta)| - M_n \,\big| = O(q)$. Combining the three edit-distance bounds gives $\operatorname{edit}(G, H_n) = O(q)$, which is the linear closeness condition. □

The unbalanced bipartite case illustrates this condition through the formula for the number of edges. If $a + b = n$, $a \leq b$, and $b - a \geq \gamma n$, then $(a + u)(b - u) - ab = u(b - a - u)$. For $|u| \geq 1$ and $|u| \leq (b - a)/2$ this has absolute value at least $\gamma n/2$. Hence an $O(q)$ error in the number of edges with $q \leq \varepsilon n$ forces $u = 0$ after ε is chosen small. The one-large-part split graph has an analogous linear calculation, since $\binom{n}{2} - \binom{a+u}{2} - (\binom{n}{2} - \binom{a}{2}) = -ua - \binom{u}{2}$. When $a \geq \gamma n$, any non-zero u changes the number of edges by $\Omega(n)$.

6.3 Quadratic splitting

The next case occurs when secondary symmetric parts have to be split into singleton parts. The quadratic effect is already visible in the simplest three-part model for $K_{\ell,1,1}$. In a complete tripartite graph with part sizes x, y, z, the number of induced copies of $K_{\ell,1,1}$ is

$$\Psi_\ell(x,y,z) \coloneqq \binom{x}{\ell}yz + \binom{y}{\ell}xz + \binom{z}{\ell}xy. \tag{6.1}$$

If x is linear-sized and $y, z = o(x)$, then along the symmetric perturbation $(x, b+t, b-t)$,

$$\Psi_\ell(x, b+t, b-t) = \Psi_\ell(x,b,b) - \binom{x}{\ell}t^2 + O_\ell(xb^{\ell-1}t^2 + xb^{\ell-2}|t|^3). \tag{6.2}$$

Thus the leading loss is quadratic in the balancing parameter. In the actual extremal construction for $K_{\ell,1,1}$ the secondary vertices form a clique, equivalently a set of singleton parts. A secondary part of size $u \geq 2$ loses the same kind of quadratic amount, because the two singleton vertices of $K_{\ell,1,1}$ can no longer both be chosen inside that part.

The following estimate is the quadratic reduction needed for the split-graph application. It says that each secondary non-singleton pair has a definite copy-count incentive to split into singleton parts.

Proposition 6.3. *Fix $\ell \geq 3$. Let B be a complete-partite n-vertex graph with a distinguished part A of size $a \geq \alpha n$, where $\alpha > 0$ is fixed, and suppose every other non-singleton part has size at most ηn. Let these other non-singleton parts have sizes $u_1, \dots, u_s$, and put the* quadratic splitting defect $Q(B) \coloneqq \sum_{i\in[s]} \binom{u_i}{2}$. *Let B^* be obtained from B by splitting all these parts into singleton parts. If η is sufficiently small in terms of α and ℓ, then*

$$N(K_{\ell,1,1}, B^*) - N(K_{\ell,1,1}, B) \geq c_{\ell,\alpha} Q(B) n^\ell.$$

Moreover $|B^*| - |B| = Q(B)$ *and* $\operatorname{edit}(B, B^*) = Q(B)$.

Proof. It is enough to split the secondary non-singleton parts one at a time. Fix such a part U of size $u \geq 2$. Splitting U into u singleton parts adds exactly the $\binom{u}{2}$ pairs inside U and changes no other adjacency. Hence the edge increase and the edit-distance contribution are both $\binom{u}{2}$. The split creates at least $\binom{u}{2}\binom{a}{\ell}$ copies: choose two vertices of U as the two singleton vertices of $K_{\ell,1,1}$, and choose the ℓ-vertex independent part inside the distinguished part A. These copies did not exist before the split, because two vertices in U were non-adjacent. The only copies that can be destroyed contain at least one of the newly added pairs inside U. In an induced $K_{\ell,1,1}$, the only non-edges are inside its ℓ-vertex independent part. Since a complete-partite host has non-edges only within host parts, any destroyed copy using a newly added pair from U must have its whole ℓ-vertex independent class inside U. The two singleton vertices are then chosen outside U, in at most $\binom{n}{2}$ ways. Hence the number of destroyed copies is at most $\binom{u}{\ell}\binom{n}{2}$. Since $a \geq \alpha n$, there is $c_1 = c_1(\ell, \alpha) > 0$ such that $\binom{a}{\ell} \geq c_1 n^\ell$ for all sufficiently large n. Also, if $u \leq \eta n$, then $\binom{u}{\ell}\binom{n}{2} \leq C_\ell u^\ell n^2 = C_\ell (u/n)^{\ell-2} u^2 n^\ell \leq C_\ell \eta^{\ell-2} u^2 n^\ell$. Choosing η sufficiently small in terms of ℓ and α, we obtain $\binom{u}{2}\binom{a}{\ell} - \binom{u}{\ell}\binom{n}{2} \geq c_{\ell,\alpha}\binom{u}{2}n^\ell$. Summing over all secondary non-singleton parts gives the desired gain. The formulae $|B^*| - |B| = Q(B)$ and $\operatorname{edit}(B, B^*) = Q(B)$ follow from the fact that the split adds exactly the internal pairs of the secondary parts and changes no other adjacencies. ☐

As an immediate consequence, a near-optimal complete-partite graph in this regime is already close to a split graph.

Corollary 6.4. *Fix $\ell \geq 3$. Suppose B is a complete-partite n-vertex graph satisfying $N(K_{\ell,1,1}, B) \geq I(K_{\ell,1,1}, n) - Cqn^\ell$, and suppose that its largest part has size at least αn, while all other non-singleton parts have size at most ηn with η as in Proposition 6.3. Then B is within $O_{\ell,\alpha,C}(q)$ edge edits of a split graph.*

Proof. Let B^* be the split graph obtained by splitting all secondary non-singleton parts. Since B^* is an n-vertex graph, $N(K_{\ell,1,1}, B^*) \leq I(K_{\ell,1,1}, n)$. Proposition 6.3 gives $c_{\ell,\alpha}Q(B)n^\ell \leq N(K_{\ell,1,1}, B^*) - N(K_{\ell,1,1}, B) \leq Cqn^\ell$. Thus $Q(B) = O(q)$, and splitting the secondary parts requires exactly $Q(B) = O(q)$ edge edits. ☐

The map $B \mapsto B^*$ therefore has edit cost $Q(B)$, changes the number of edges by $Q(B)$, and increases the number of copies by $\Omega(Q(B)n^\ell)$. After the secondary parts have been split, the remaining split family is controlled by the one-large-part calculation from Section 6.2.

6.4 Finite complete-bipartite estimates

This subsection gives the finite estimates used for the $K_{s,t}$ application. The external inputs are Brown–Sidorenko symmetrization [4, Propositions 1–2] and the complete-bipartite perfect-stability theorem of Liu, Pikhurko, Sharifzadeh, and Staden [11, Theorem 1.3]. The finite point is that, in the unbalanced regime, the exact bipartite optimizer is fixed by the number of edges.

We use Brown–Sidorenko in the following specific form: after symmetrization, the limiting $K_{s,t}$ inducibility problem is reduced to the one-variable function

$$f_{s,t}(x) \coloneqq x^s(1-x)^t + x^t(1-x)^s, \qquad 1/2 \le x \le 1,$$

and its maximizing ratio in $[1/2, 1]$ is unique by [11, Theorem 1.3]. The calculation below only shows that $1/2$ is not a local maximum.

Let $1 \le s < t$, put $k \coloneqq s+t$, and define $f_{s,t}(x) \coloneqq x^s(1-x)^t + x^t(1-x)^s$ for $x \in [1/2, 1]$.

Lemma 6.5. *If $s < \binom{t-s}{2}$, then the unique maximizer $\alpha \in [1/2, 1]$ of $f_{s,t}$ satisfies $\alpha > 1/2$. Consequently there is $\gamma = \gamma(s,t) > 0$ such that every finite bipartite maximizer K_{a_n,b_n}, with $a_n \le b_n$, satisfies $b_n - a_n \ge \gamma n$ for all sufficiently large n.*

Proof. A direct calculation gives $f''_{s,t}(1/2) = 2^{3-k}((t-s)^2 - (s+t)) = 2^{3-k}((t-s)(t-s-1) - 2s)$. The condition $s < \binom{t-s}{2}$ makes this positive, so $1/2$ is not a local maximum. The uniqueness assertion is [11, Theorem 1.3]; hence its maximizer α satisfies $\alpha > 1/2$. If $b_n - a_n = o(n)$ for a subsequence of finite bipartite maximizers, then their part ratios have a further subsequence converging to $1/2$. Dividing $\binom{x}{s}\binom{n-x}{t} + \binom{x}{t}\binom{n-x}{s}$ by n^k and passing to the limit would make $1/2$ a limiting maximizer of $f_{s,t}$, a contradiction. ☐

Near this unbalanced optimum, any residual part outside the two main parts can be eliminated with a linear gain in the number of moved vertices.

Lemma 6.6. *Let $1 \le s < t$ and put $k \coloneqq s+t$. For every $c_0 > 0$ there are $\rho > 0$ and $c > 0$ with the following property. Let C be a complete multipartite n-vertex graph with two distinguished parts A, B satisfying $|A|, |B| \ge c_0 n$, and let R be the* residual set*, namely the union of all remaining parts. If $|R| \le \rho n$, then replacing the vertices of R, one at a time, by the better clone in A or B gives a complete bipartite graph C' such that*

$$N(K_{s,t}, C') - N(K_{s,t}, C) \ge c|R|n^{k-1}$$

and $\mathrm{edit}(C, C') \le |R|n$.

Proof. Move the vertices of R one at a time. Consider one such vertex v at an intermediate graph, and write $a \coloneqq |A|$ and $b \coloneqq |B|$, relabeling A, B if necessary so that $b \ge a$. Since at most $|R| \le \rho n$ vertices have been moved, $a, b \ge c_0 n/2$ after decreasing ρ. If v is made an A-clone, then using only $A \cup B \cup \{v\}$ and requiring the copy to contain v gives at least $R_A(a,b) \coloneqq \binom{a}{s-1}\binom{b}{t} + \binom{a}{t-1}\binom{b}{s}$ copies. If v is made a B-clone, the analogous number is $R_B(a,b) \coloneqq \binom{b}{s-1}\binom{a}{t} + \binom{b}{t-1}\binom{a}{s}$. When $s \ge 2$, $\max\{R_A(a,b), R_B(a,b)\} \ge c_1 n^{k-1}$, while before the move there is no copy containing v and otherwise only vertices of $A \cup B$, because v is adjacent to both A and B and cannot join either non-singleton independent class. Copies containing v and at least one other residual vertex contribute at most $C_1|R|n^{k-2} \le C_1 \rho n^{k-1}$ to the comparison. Choosing ρ small gives a gain of at least $c_2 n^{k-1}$ for this move.

It remains to handle $s = 1$. Before the move, the copies containing v and otherwise only vertices of $A \cup B$ are at most $\binom{a}{t} + \binom{b}{t}$. Making v an A-clone gives $\binom{b}{t} + b\binom{a}{t-1}$ such copies. Hence the gain from these copies is at least $b\binom{a}{t-1} - \binom{a}{t} = \binom{a}{t-1}(b - (a-t+1)/t)$, which is $\Omega_t(n^t) = \Omega_t(n^{k-1})$ because $a, b \ge c_0 n/2$. Again the contribution of copies using another residual vertex is $O_t(|R|n^{k-2})$, and it is absorbed by taking ρ small.

Thus every moved residual vertex increases the number of induced $K_{s,t}$'s by at least $c_3 n^{k-1}$. Summing over all vertices of R gives the asserted gain. Each moved vertex changes at most n incident adjacencies, so $\mathrm{edit}(C, C') \le |R|n$. ☐

Combining the residual cleanup with the Brown–Sidorenko limiting optimizer gives the finite exact template.

Lemma 6.7. *Assume $1 \le s < t$ and $s < \binom{t-s}{2}$. For all sufficiently large n, every exact n-vertex extremal graph for $K_{s,t}$ is a complete bipartite graph $K_{a,b}$ maximizing $\binom{a}{s}\binom{b}{t} + \binom{a}{t}\binom{b}{s}$ over $a + b = n$.*

Proof. Brown–Sidorenko symmetrization [4, Proposition 1] shows that an extremal graph may be chosen complete multipartite, and perfect stability [11, Theorem 1.3] implies that every exact extremal graph is complete multipartite. Apply Lemma 6.1 to the unique limiting maximizer $(\alpha, 1-\alpha, 0, \ldots)$ from Lemma 6.5. It follows that every such extremal graph has two linear-sized parts A, B, while the union R of all remaining parts has size $o(n)$. If R were non-empty, Lemma 6.6 would replace its vertices by clones in $A \cup B$ and strictly increase the number of induced $K_{s,t}$'s for all sufficiently large n, contradicting exact extremality. Hence no residual part exists, and the graph is complete bipartite. Among complete bipartite graphs, the displayed polynomial is exactly the number of induced $K_{s,t}$'s, so the finite exact extremal graphs are precisely its maximizers. □

We also need the local pair-flip losses around this bipartite template. The formulas below give both strictness and the correct order of the loss.

Lemma 6.8. *Let $H := K_{a,b}$ with $a, b = \Theta(n)$, and let A, B be its two parts. If $xy \subseteq A$, then adding xy destroys $\binom{a-2}{s-2}\binom{b}{t} + \binom{a-2}{t-2}\binom{b}{s}$ induced copies of $K_{s,t}$, with the convention $\binom{u}{v} = 0$ for $v < 0$. If $xy \subseteq B$, the analogous loss is $\binom{b-2}{s-2}\binom{a}{t} + \binom{b-2}{t-2}\binom{a}{s}$. If $x \in A$ and $y \in B$, then deleting xy destroys $\binom{a-1}{s-1}\binom{b-1}{t-1} + \binom{a-1}{t-1}\binom{b-1}{s-1}$ induced copies. In the unbalanced optimizer neighborhood all away flips are strict and have loss $\Omega_{s,t}(n^{k-2})$.*

Proof. For an internal pair $xy \subseteq A$, an induced $K_{s,t}$ is destroyed exactly when xy lies in the independent class placed in A. This gives the two terms according to whether the s-class or the t-class is placed in A. No copy is created by adding an internal edge in a complete bipartite graph. The formula for $xy \subseteq B$ is symmetric. For a cross pair xy, a copy is destroyed exactly when x and y lie in the two different independent classes of the $K_{s,t}$, giving the displayed two terms. Deleting a cross edge creates no new induced $K_{s,t}$. Since $a, b = \Theta(n)$, the displayed quantities have order n^{k-2} whenever they are evaluated in the unbalanced optimizer neighborhood, and the positive lower bound is uniform there. □

The next lemma is the finite fixed-edge input: near optimality and the prescribed edge count force $O(q)$ closeness to the exact bipartite graph.

Lemma 6.9. *Assume $1 \le s < t$ and $s < \binom{t-s}{2}$. Let $H_n^{s,t} := K_{a_n,b_n}$ be a finite exact bipartite maximizer with $a_n \le b_n$, and put $M_n^{s,t} := |H_n^{s,t}| = a_n b_n$. For every $C_0 > 0$, there are $C_R, \varepsilon_R > 0$ such that if $1 \le q \le \varepsilon_R n$, $|G| = M_n^{s,t} + q$, and $N(K_{s,t}, G) \ge N(K_{s,t}, H_n^{s,t}) - C_0 q n^{k-2}$, then* $\mathrm{edit}(G, H_n^{s,t}) \le C_R q$.

Proof. Perfect stability gives a complete multipartite graph C with $\mathrm{edit}(G, C) = O(q)$. Changing $O(q)$ adjacencies changes the number of induced $K_{s,t}$ copies by $O(qn^{k-2})$, so $N(K_{s,t}, C) \ge I(K_{s,t}, n) - O(qn^{k-2})$ and $\big| |C| - |G| \big| = O(q)$. Since $q = O(n)$, Lemma 6.1 puts $x(C)$ in any prescribed neighborhood of $(\alpha, 1-\alpha, 0, \ldots)$ once n is large. In particular, C has two linear-sized parts A, B and residual set R with $|R| \le \rho n$, where ρ is the constant in Lemma 6.6. Let C' be the complete bipartite graph obtained by eliminating R as in that lemma. Since C' is an n-vertex graph, $N(K_{s,t}, C') \le I(K_{s,t}, n)$. Hence $c|R|n^{k-1} \le N(K_{s,t}, C') - N(K_{s,t}, C) \le O(qn^{k-2})$, and therefore $|R|n = O(q)$. Lemma 6.6 gives $\mathrm{edit}(C, C') = O(q)$. Write $C' := K_{x,n-x}$ with $x \le n/2$. Since $|G| = a_n b_n + q$ and G is $O(q)$ edits from C', we have $|x(n-x) - a_n b_n| = O(q)$. By Lemma 6.5, $b_n - a_n \ge \gamma n$, and finite localization also gives $x/n = a_n/n + o(1)$. Thus $|x(n-x) - a_n(n-a_n)| \ge cn|x - a_n|$, so $n|x - a_n| = O(q)$. The edit distance from $K_{x,n-x}$ to $H_n^{s,t}$ is $O(n|x - a_n|) = O(q)$, and the triangle inequality proves the claim. □

We package the preceding estimates in the form needed by the general fixed-edge theorem.

Proposition 6.10. *Let $1 \le s < t$ and assume $s < \binom{t-s}{2}$. Let $k := s + t$. There are finite exact maximizers $H_n^{s,t} := K_{a_n,b_n}$, with $a_n \le b_n$, such that $b_n - a_n \ge \gamma n$ for some $\gamma = \gamma(s,t) > 0$ and all sufficiently large n. The inducibility problem for $K_{s,t}$ is perfectly stable, every near-optimal graph with $|G| = |H_n^{s,t}| + q$, $1 \le q \le \varepsilon n$, is $O(q)$ edits from $H_n^{s,t}$, and $H_n^{s,t}$ is pair-flip strict with loss $\Omega_{s,t}(n^{k-2})$.*

Proof. The separation $b_n - a_n \ge \gamma n$ is Lemma 6.5. Perfect stability is the theorem of Liu, Pikhurko, Sharifzadeh, and Staden for complete bipartite inducibility [11, Theorem 1.3]. The finite exact extremal statement is Lemma 6.7. Pair-flip strictness is Lemma 6.8. The $O(q)$ reduction to the exact bipartite graph in the small linear range above $M_n^{s,t}$ is Lemma 6.9. □

6.5 A cubic moment inequality for the split copy count

The limiting calculation for $K_{\ell,1,1}$ uses one elementary inequality when all non-singleton parts have mass at most $1/2$.

Lemma 6.11. *Let $0 \le x_i \le 1/2$, $\sum_i x_i \le 1$, and $S_j := \sum_i x_i^j$. Then $(1-S_2)S_3 - 2S_4 + 2S_5 \le 1/32$.*

Proof. By truncation and compactness, it is enough to prove the inequality for finitely many positive coordinates. Put $y := S_2$, $z := S_3$, and $C := \sum_i x_i^4(1-x_i)$. Then $(1-S_2)S_3 - 2S_4 + 2S_5 = (1-y)z - 2C$. By Cauchy's inequality,

$$z^2 = \left(\sum_i x_i^3\right)^2 = \left(\sum_i x_i^2\sqrt{1-x_i} \cdot \frac{x_i}{\sqrt{1-x_i}}\right)^2 \le C \sum_i \frac{x_i^2}{1-x_i}.$$

Since $0 \le x_i \le 1/2$, we have $(1-x_i)^{-1} \le 1 + 2x_i$, and hence $\sum_i x_i^2/(1-x_i) \le y + 2z$. Thus $C \ge z^2/(y+2z)$, with the conclusion trivial if $y = z = 0$. Assume now that $y > 0$, and put $r := z/y$. Then $0 \le r \le 1/2$, because r is a weighted average of the x_i with weights x_i^2/y. Therefore

$$(1-y)z - 2C \le (1-y)z - \frac{2z^2}{y+2z} = yr\left(\frac{1}{1+2r} - y\right).$$

For fixed r, writing $A := (1+2r)^{-1}$, we have $y(A-y) \le A^2/4$. Consequently $(1-y)z - 2C \le r/(4(1+2r)^2)$. The function $r/(4(1+2r)^2)$ is increasing on $[0,1/2]$, since its derivative is $(1-2r)/(4(1+2r)^3)$. Hence $(1-S_2)S_3 - 2S_4 + 2S_5 \le 1/32$. □

6.6 Finite split-graph estimates for $K_{\ell,1,1}$

Fix $\ell \ge 3$. In the complete-partite limit notation, let $x = (x_1, x_2, \ldots)$, $x_0 := 1 - \sum_{i\ge1} x_i$, and $Q := \sum_{i\ge1} x_i^2$. For $K_{\ell,1,1}$, up to a positive constant depending only on ℓ, the complete-partite limit copy count is $T_\ell(x) := \sum_{i\ge1} x_i^\ell B_i$, where $B_i := 1 - Q - 2x_i + 2x_i^2$. Here B_i is the ordered density of adjacent pairs outside part i. The first step is to identify the unique limiting split optimizer.

Lemma 6.12. *For every $\ell \ge 3$,*

$$T_\ell(x) \le \left(\frac{\ell}{\ell+2}\right)^\ell \left(\frac{2}{\ell+2}\right)^2,$$

with equality if and only if $x = (\ell/(\ell+2), 0, 0, \ldots)$ and $x_0 = 2/(\ell+2)$.

Proof. Let $m := \max_i x_i$, and assume $x_1 = m$. Put $R := 1 - m$ and $q_2 := \sum_{i\ge2} x_i^2$. Then $B_1 = R^2 - q_2$. For $i \ge 2$, we have $B_i \le (R - x_i)(1 + m - x_i)$. If $m \ge 1/2$, then $x_i B_i \le x_i(R-x_i)(1+m-x_i) \le R^2(1+m)/4 \le m^3$. Hence $x_i^3 B_i \le m^3 x_i^2$ for $i \ge 2$, and therefore $T_3(x) = m^3(R^2 - q_2) + \sum_{i\ge2} x_i^3 B_i \le m^3 R^2$. For $\ell = 3$ this gives $T_3(x) \le m^3(1-m)^2$. For $\ell \ge 4$, since $x_i \le m$, we get $T_\ell(x) = \sum_i x_i^{\ell-3} x_i^3 B_i \le m^{\ell-3} T_3(x) \le m^\ell(1-m)^2$. Thus, for every $\ell \ge 3$ in the case $m \ge 1/2$, $T_\ell(x) \le m^\ell(1-m)^2$. The one-variable function $m^\ell(1-m)^2$ is uniquely maximized on $[0,1]$ at $m = \ell/(\ell+2)$, with maximum $(\ell/(\ell+2))^\ell(2/(\ell+2))^2$. Equality forces $q_2 = 0$, so all other positive non-singleton parts vanish.

It remains to treat $m < 1/2$. This is exactly Lemma 6.13. Thus no optimizer has $m < 1/2$, and the equality case is the one already described. □

It remains to supply the sub-half estimate used in the preceding proof. The cubic inequality handles $\ell = 3$, and higher ℓ follows by comparison with the cubic case.

Lemma 6.13. *Let $\ell \ge 3$, and let $x = (x_1, x_2, \ldots) \in \mathcal{P}$. With $T_\ell(x) := \sum_i x_i^\ell(1 - Q - 2x_i + 2x_i^2)$, where $Q := \sum_i x_i^2$, suppose that $\max_i x_i \le 1/2$. Then $T_\ell(x) < (\ell/(\ell+2))^\ell(2/(\ell+2))^2$.*

Proof. It is enough to prove the estimate for finite support and then pass to the limit by truncation. For $\ell = 3$, write $S_j := \sum_i x_i^j$. Then $T_3(x) = (1-S_2)S_3 - 2S_4 + 2S_5$, and Lemma 6.11 gives $T_3(x) \le 1/32$. Since $1/32 < (3/5)^3(2/5)^2$, the result follows for $\ell = 3$.

Assume $\ell \ge 4$. For every i, the quantity $B_i = 1 - Q - 2x_i + 2x_i^2$ is the ordered density of adjacent pairs outside the i-th non-singleton part, and in particular $B_i \ge 0$; also $B_i \le 1$. Since $x_i \le 1/2$, we have $x_i^\ell \le 2^{3-\ell} x_i^3$. Hence $T_\ell(x) = \sum_i x_i^\ell B_i \le 2^{3-\ell} \sum_i x_i^3 B_i = 2^{3-\ell} T_3(x) \le 2^{-\ell-2}$. It remains to check that $2^{-\ell-2} < (\ell/(\ell+2))^\ell(2/(\ell+2))^2$ for every $\ell \ge 4$. Equivalently, $R_\ell := ((\ell+2)^{\ell+2})/(2^{\ell+4}\ell^\ell) < 1$. Directly $R_4 = 729/1024 < 1$. For $\ell \ge 4$,

$$\frac{R_{\ell+1}}{R_\ell} = \frac{1}{2}\left(\frac{\ell}{\ell+1}\right)^\ell \left(\frac{\ell+3}{\ell+2}\right)^{\ell+2} \frac{\ell+3}{\ell+1} < \frac{1}{2} e^{-\ell/(\ell+1)} e \frac{\ell+3}{\ell+1} = \frac{1}{2} e^{1/(\ell+1)} \frac{\ell+3}{\ell+1} < 1.$$

Thus $R_\ell \le R_4 < 1$ for all $\ell \ge 4$, proving the strict inequality. □

We next record the exact one-pair losses in a finite split graph. These formulas provide the pair-flip strictness needed by the general reduction.

Lemma 6.14. *Let $H := S_{a,b}$ with $a, b = \Theta(n)$, where A is the independent part of size a and B is the clique of size b. Every pair flip away from the split graph decreases $N(K_{\ell,1,1}, \cdot)$ by $\Omega_\ell(n^\ell)$. More precisely, $\Delta(H, K_{\ell,1,1}; xy) = \binom{a-2}{\ell-2}\binom{b}{2}$ if $xy \subseteq A$, $\Delta(H, K_{\ell,1,1}; xy) = \binom{a}{\ell}$ if $xy \subseteq B$, and $\Delta(H, K_{\ell,1,1}; xy) = \binom{a-1}{\ell-1}(b-1)$ if $x \in A$ and $y \in B$.*

Proof. If $xy \subseteq A$, then adding xy destroys exactly the copies in which x, y lie in the ℓ-vertex independent class and the two singleton vertices lie in B, giving $\binom{a-2}{\ell-2}\binom{b}{2}$. No copy is created: x, y cannot serve as the two singleton vertices, because vertices of A are not adjacent to the rest of the independent part. If $xy \subseteq B$, then deleting xy destroys exactly the copies in which x, y are the two singleton vertices and the ℓ-vertex part is chosen in A, namely $\binom{a}{\ell}$. No copy is created, since xy cannot become part of the ℓ-vertex independent class when $\ell \geq 3$. Finally, if $x \in A$ and $y \in B$, deleting xy destroys exactly the copies in which x lies in the ℓ-vertex independent class, y is one singleton vertex, and the other singleton vertex is chosen in $B \setminus \{y\}$. This gives $\binom{a-1}{\ell-1}(b-1)$. Again no copy is created, because after deleting one cross edge there is no third vertex non-adjacent to both x and y which could complete an ℓ-vertex independent class. Since $a, b = \Theta(n)$, all displayed quantities are $\Theta_\ell(n^\ell)$. ☐

To apply the symmetrization-stability criterion, pair-flip strictness is not enough; we also need the rooted one-vertex strictness condition.

Lemma 6.15. *The limiting split optimizer $x_\ell := (\ell/(\ell+2), 0, 0, \ldots)$, with singleton mass $2/(\ell+2)$, satisfies the* vertex-cloning strictness *inequality in the criterion of Liu, Pikhurko, Sharifzadeh, and Staden [11, Theorem 1.1].*

Proof. We check the rooted one-vertex inequality in the limiting split model; this avoids any finite estimate comparing the two clone values. Put $\alpha := \ell/(\ell+2)$ and $\beta := 2/(\ell+2)$. Let $S_{a,b}$ be a split graph with independent part A and clique part B, where $a/n \to \alpha$ and $b/n \to \beta$. Consider a new vertex v with arbitrary attachment to $A \cup B$, and write $x := |N(v) \cap A|$, $z := |N(v) \cap B|$, $u := x/a$, and $w := z/b$. The number of induced copies of $K_{\ell,1,1}$ containing v is

$$R(x, z) := \binom{a-x}{\ell-1}\binom{z}{2} + \binom{x}{\ell}z.$$

Indeed, in the first term v lies in the ℓ-vertex independent class, and in the second term v is one of the two singleton vertices. There are no other cases because $\ell \geq 3$, the clique part contains no two non-adjacent vertices, and every vertex of B is adjacent to every vertex of A.

The two clone attachments are the A-clone $(x, z) = (0, b)$ and the clique-clone $(x, z) = (a, b)$. Their rooted leading constants are equal in the limit:

$$R_\star := \frac{\alpha^{\ell-1}\beta^2}{2(\ell-1)!} = \frac{\alpha^\ell \beta}{\ell!}.$$

Moreover, uniformly for $0 \leq u, w \leq 1$,

$$\frac{R(x, z)}{n^{\ell+1}} = R_\star\big((1-u)^{\ell-1}w^2 + u^\ell w\big) + o(1).$$

Thus the limiting rooted loss from replacing v by either clone, normalized by $R_\star n^{\ell+1}$, is $g(u, w) := 1 - (1-u)^{\ell-1}w^2 - u^\ell w$.

We claim that there is $c_\ell > 0$ such that $g(u, w) \geq c_\ell(\min\{u, 1-u\} + 1 - w)$ for all $0 \leq u, w \leq 1$. First, $g(u, 1) = 1 - (1-u)^{\ell-1} - u^\ell$ is positive on $(0, 1)$ and has positive one-sided linear slopes at 0 and 1, so compactness gives $g(u, 1) \geq c_1 \min\{u, 1-u\}$. Second, $g(u, w) - g(u, 1) = (1-u)^{\ell-1}(1-w^2) + u^\ell(1-w)$, and the coefficient of $1 - w$ in this expression is $(1-u)^{\ell-1}(1+w) + u^\ell$, which is bounded below by a positive constant depending only on ℓ. The claim follows.

The normalized incident edit distance from the attachment of v to the nearest clone is comparable, with constants depending only on ℓ, to $\min\{\alpha u + \beta(1-w), \alpha(1-u) + \beta(1-w)\}$, and hence to $\min\{u, 1-u\} + 1 - w$. Since $R_\star > 0$, the preceding inequality says exactly that the limiting rooted loss from replacing v by a clone is bounded below by a positive constant times the normalized incident edit distance from the clone set. This is the *vertex-cloning strictness* condition required by the symmetrization-stability criterion of Liu, Pikhurko, Sharifzadeh, and Staden. ☐

The preceding limit, pair-flip, and cloning estimates identify the exact finite extremal graphs.

Proposition 6.16. *For every fixed $\ell \geq 3$, the inducibility optimizer for $K_{\ell,1,1}$ has, in the limit, one non-singleton part of mass $\ell/(\ell+2)$ and singleton parts of total mass $2/(\ell+2)$. The inducibility problem is perfectly stable. Moreover, for all sufficiently large n, every exact n-vertex extremal graph is a split graph $S_{a,b}$ maximizing $\binom{a}{\ell}\binom{b}{2}$ over $a+b=n$.*

Proof. We verify the hypotheses of the Liu–Pikhurko–Sharifzadeh–Staden criterion. The target $K_{\ell,1,1}$ is complete multipartite, so the inducibility problem is symmetrizable in their sense. The unique limiting optimizer is the split vector from Lemma 6.12: it has one non-singleton part of mass $\ell/(\ell+2)$ and singleton mass $2/(\ell+2)$, represented in finite graphs by singleton parts. Pair-flip strictness is Lemma 6.14, and vertex-cloning strictness for non-clone one-vertex extensions is Lemma 6.15. Thus the hypotheses of Theorem 6.2 are verified, and the inducibility problem for $K_{\ell,1,1}$ is perfectly stable.

We now identify the finite exact extremal graphs. The unnormalized perfect-stability bound implies that, for all sufficiently large n, every exact extremal graph has zero edit distance from the relevant complete-partite family, and hence is complete multipartite. By Lemma 6.1, such a graph has one part of size $(\ell/(\ell+2)+o(1))n$, while every other non-singleton part has size $o(n)$. Let U be a secondary non-singleton part of size $u \geq 2$, and let A be the largest part, with $|A| \geq c_\ell n$. Split U into u singletons. This creates at least $\binom{u}{2}\binom{|A|}{\ell}$ copies in which two vertices of U are the singleton vertices and the ℓ-vertex part lies in A. The only possible destroyed copies are those in which the ℓ-vertex part was chosen inside U; their number is at most $\binom{u}{\ell}\binom{n}{2}$. Since $u = o(n)$ and $\ell \geq 3$, $\binom{u}{2}\binom{|A|}{\ell} - \binom{u}{\ell}\binom{n}{2} > 0$ for all sufficiently large n. Hence no secondary non-singleton part can occur in a finite exact extremal graph. Therefore every exact extremal graph is split. For a split graph $S_{a,b}$, every induced $K_{\ell,1,1}$ chooses its ℓ-vertex independent class in the independent part and the two singleton vertices in the clique. Hence $N(K_{\ell,1,1}, S_{a,b}) = \binom{a}{\ell}\binom{b}{2}$. Thus the finite exact extremal graphs are precisely the split graphs maximizing this expression. □

Finally we need the fixed-edge closeness input for the split exact graph. Quadratic splitting removes all secondary non-singleton parts, and the edge count then fixes the remaining split parameter linearly.

Proposition 6.17. *Fix $\ell \geq 3$. Let $H_n^\ell := S_{a_n,b_n}$ be the exact split graph from Proposition 6.16, and put $M_n^\ell := |H_n^\ell|$. For every $C_0 > 0$, there are $C_R > 0$, $\varepsilon_R > 0$, and n_0 such that the following holds for all $n \geq n_0$. If $1 \leq q \leq \varepsilon_R n$ and G satisfies $|G| = M_n^\ell + q$ and $N(K_{\ell,1,1}, G) \geq N(K_{\ell,1,1}, H_n^\ell) - C_0 q n^\ell$, then $\operatorname{edit}(G, H_n^\ell) \leq C_R q$.*

Proof. Perfect stability gives a complete-partite graph B with $\operatorname{edit}(G,B) = O(q)$. Since changing $O(q)$ adjacencies changes the number of induced $K_{\ell,1,1}$ copies by $O_\ell(qn^\ell)$, the hypothesis on G implies $N(K_{\ell,1,1}, B) \geq I(K_{\ell,1,1}, n) - O_\ell(qn^\ell)$. Since $q = O(n)$, Lemma 6.1, applied to the unique optimizer from Lemma 6.12, shows that B has one non-singleton part of size at least αn and all other non-singleton parts have size at most ηn, where η is small enough for Proposition 6.3. Let B^* be the split graph obtained by splitting all secondary non-singleton parts, and let $Q(B)$ be the corresponding quadratic splitting defect. Then $cQ(B)n^\ell \leq N(K_{\ell,1,1}, B^*) - N(K_{\ell,1,1}, B) \leq I(K_{\ell,1,1}, n) - N(K_{\ell,1,1}, B) = O(qn^\ell)$, and so $Q(B) = O(q)$. Hence $\operatorname{edit}(B, B^*) = O(q)$ and $\big||B^*| - |B|\big| = O(q)$. Write $B^* = S_{x,n-x}$, where x is the size of its independent part. Since G, B, and B^* differ by $O(q)$ adjacencies in total, $\big||S_{x,n-x}| - |S_{a_n,b_n}|\big| = O(q)$. But $|S_{x,n-x}| - |S_{a_n,b_n}| = -(\binom{x}{2} - \binom{a_n}{2})$, and finite localization gives $x/n, a_n/n \to \ell/(\ell+2)$; hence $|\binom{x}{2} - \binom{a_n}{2}| \geq c_\ell n|x - a_n|$ whenever $x \neq a_n$ and n is large. Thus $n|x - a_n| = O(q)$, so $\operatorname{edit}(B^*, H_n^\ell) = O(q)$. The triangle inequality gives $\operatorname{edit}(G, H_n^\ell) = O(q)$. □

6.7 Proofs of the application theorems

We finish the section by applying the preceding finite estimates to the two application statements from Section 2.2.

6.7.1 Unbalanced complete bipartite graphs

Proof of Theorem 2.5. We use the notation from Theorem 2.5. Proposition 6.10 gives the three estimates needed for the small linear range form of Theorem 2.3: exact finite extremal graphs with $M_n^{s,t}$ edges, the linear closeness condition for graphs with $M_n^{s,t} + q$ edges, and pair-flip strictness at $H_n^{s,t}$. Indeed, suppose G is a graph with $|G| = M_n^{s,t} + q$ and $N(K_{s,t}, G) \geq N(K_{s,t}, H_n^{s,t}) - O_{s,t}(qn^{k-2})$. By Proposition 6.10, G is $O(q)$ edits from $H_n^{s,t}$, provided $q \leq \varepsilon n$ and ε is sufficiently small. Lemma 6.8 gives pair-flip strictness with loss $\Omega_{s,t}(n^{k-2})$. Therefore Proposition 3.4, equivalently Theorem 2.3, applies and gives the stated formula and equality characterisation. □

6.7.2 The split graphs $K_{\ell,1,1}$

Proof of Theorem 2.6. Use the notation from Theorem 2.6, and write $H_n^\ell := S_{a_n,b_n}$ with independent part A and clique B. Proposition 6.16 gives the exact split graph and perfect stability. Proposition 6.17 gives the finite $O(q)$ reduction from a near-optimal complete-partite graph to the exact split graph. Lemma 6.14 gives pair-flip strictness at H_n^ℓ. Therefore Theorem 2.3, in its small linear range form, gives the exact value and equality characterisation.

Finally, adding a pair $xy \subseteq A$ destroys exactly the copies in which x, y lie in the ℓ-vertex independent class and the two singleton vertices lie in the clique B, namely $\binom{a_n-2}{\ell-2}\binom{b_n}{2}$. The displayed first-order expansion follows from Lemma 3.2, since here $v(K_{\ell,1,1}) = \ell + 2$. □

7 Concluding remarks

The main theorem strengthens exact Turán inducibility by fixing the number of edges. The stability theorem moves a near maximizer to a balanced complete multipartite graph, but it does not by itself identify the optimum at each nearby value of $|G|$. The balanced part-size comparison supplies the missing edge-count comparison: an incorrect balanced complete multipartite graph has an edge deficit and must pay additional marginal losses to reach the prescribed number of edges.

The later applications show two other ways in which the number of edges can determine the finite extremal graph. For unbalanced $K_{s,t}$, the relevant part size is fixed by a linear calculation of the number of edges. For $K_{\ell,1,1}$, one first splits the secondary parts and then uses the same linear calculation inside the split family.

The complementary lower-edge version for $K_s \sqcup K_t = \overline{K_{s,t}}$ is immediate from Theorem 2.5: taking complements transforms graphs with $|\overline{K_{a_n,b_n}}| - q$ edges into graphs with $|K_{a_n,b_n}| + q$ edges, and preserves induced-copy counts under complementation of the target.

This suggests the following concrete problems.

Problem 7.1. *Can one formulate a single theorem comparing the leftover edits which treats both fixed-edge ranges $|G| = M_n + q$ and $|G| = M_n - q$, and whose equality cases are obtained by the minimum-loss additions or deletions from the same exact graph?*

Problem 7.2. *Develop a lower-edge analogue for the range $|G| = M_n - q$. Under what local hypotheses on the exact extremal graph H_n is the optimum obtained by deleting a q-set of present pairs of minimum loss, and when can changing the extremal template, or passing to complements, lead to different behaviour?*

Problem 7.3. *For non-complete Turán targets, what are the extremal graphs when q is larger than linear but still $o(n^2)$, especially in the range where changing the part sizes and adding missing pairs can contribute on the same scale?*

Problem 7.4. *For complete multipartite targets with several local directions, can one give a finite part-size test which decides whether the optimum at a fixed number of edges is controlled by a linear direction, a balanced part-size direction, a quadratic splitting direction, or a mixture of these?*

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Declaration on the use of AI

The authors used generative AI tools to assist in discussing proof strategies, checking proofs, and improving exposition. All mathematical arguments, results, and conclusions were reviewed and verified by the authors.

A Finite complete-partite cleanup

This appendix provides the finite complete-partite cleanup estimate used in Lemma 5.2. We separate the external stability inputs from the finite bookkeeping done here.

For Turán graphs, the external input is the exact theorem, perfect stability theorem, and rooted one-vertex strictness verification of Liu, Ma, and Zhu. We use [12, Theorem 1.1] for the finite exact statement that, for every sufficiently large n, the unique extremal graph for $F = T(k,r)$ is $T(n,m)$, where m is the integer maximizing $(p)_r/p^k$. We use [12, Theorem 1.5] for perfect stability. For the local cleanup of residual parts below we use, more specifically, the condition (S2) in [12, Theorem 4.1], as verified in the proof of [12, Theorem 1.5]. Thus the limiting complete-partite maximizer relevant to Appendix A is $x_\star := (1/m, \ldots, 1/m, 0, \ldots)$.

The conversion between normalized and unnormalized distance is fixed as follows. Liu, Pikhurko, Sharifzadeh, and Staden use normalized densities, for instance $\lambda(G) := N(F,G)/\binom{n}{k}$, while this paper uses the unnormalized count $N(F,G)$. If their theorem gives normalized edit distance at most $C(\lambda(n) - \lambda(G))$, then

$$\mathrm{edit}(G,C) \le \frac{Cn^2}{2} \cdot \frac{I(F,n) - N(F,G)}{\binom{n}{k}} \le C_F \frac{I(F,n) - N(F,G)}{n^{k-2}}.$$

The first finite consequence is a rooted clone gap near the balanced m-partite Turán optimizer.

Lemma A.1. *Let $F := T(k,r)$ and let $m := m(F)$. There are constants $\rho_0, c_0 > 0$ and n_0 such that the following holds for every $n \ge n_0$. Let G be a complete m-partite graph with parts $V_1, \ldots, V_m$ satisfying $\big||V_i| - n/m\big| \le \rho_0 n$ for every i. For $A \subseteq [m]$, let G_A be obtained from G by adding a new vertex v adjacent precisely to $\bigcup_{i \in A} V_i$. If $|A| \ne m-1$, then*

$$\max_{A^* \subseteq [m],\ |A^*| = m-1} N(F, G_{A^*}) - N(F, G_A) \ge c_0 n^{k-1}.$$

All copies counted in the difference contain the new vertex v.

Proof. In the balanced graph $G_{n,x_\star}$, this is exactly the unnormalized form of condition (S2) in Liu–Ma–Zhu's stability criterion [12, Theorem 4.1], whose verification is part of their proof of [12, Theorem 1.5]. Indeed, the clone attachments are precisely the sets $A^* \subseteq [m]$ with $|A^*| = m-1$. For each fixed wrong set A, choose one clone attachment $A^*(A)$ which is optimal in the balanced graph. The rooted difference $N(F, G_{A^*(A)}) - N(F, G_A)$ is, up to $O_F(n^{k-2})$ rounding terms, a homogeneous polynomial of degree $k-1$ in the part sizes. Since there are only finitely many A with $|A| \ne m-1$ and each chosen difference is positive at $(1/m, \ldots, 1/m)$, the same lower bound with a smaller constant holds throughout a sufficiently small neighborhood of this vector. □

The rooted clone estimate is used to remove the small residual parts of a near-balanced complete multipartite graph.

Lemma A.2. *Let $F := T(k,r)$ and let $m := m(F)$. Let $x_\star := (1/m, \ldots, 1/m, 0, \ldots)$ be the partite vector whose first m coordinates are $1/m$ and whose remaining coordinates are 0. For every $\zeta > 0$ there exist constants $\rho > 0$, $c > 0$, and n_0, depending only on F and ζ, with the following property. Let C be a complete multipartite n-vertex graph with $n \ge n_0$, and suppose that the complete-partite edit distance between $x(C)$ and $x_\star$ is at most ρ. Then there is a complete m-partite graph $B := K_{b_1,\ldots,b_m}$ such that $|b_i - n/m| \le \zeta n$ for every i and*

$$N(F,B) - N(F,C) \ge c\,\mathrm{edit}(C,B) n^{k-2}.$$

Proof. Let $V_1, \ldots, V_m$ be the m largest parts of C, after relabeling, and let R be the *residual set*, namely the union of all remaining parts. If ρ is sufficiently small, then $\big||V_i| - n/m\big| \le \zeta n/4$ for every i and $|R| \le \zeta n/4$. We move the vertices of R one by one into the classes $V_1, \ldots, V_m$. Consider a vertex $v \in R$ at an intermediate graph. Its attachment to the m large classes is complete to all of them, up to the already moved $O(|R|)$ vertices. This attachment is not a clone attachment of the m-partite optimizer $x_\star$, because an optimizer clone must be non-adjacent to exactly one of the m large classes and adjacent to the other $m-1$. Since each large class has size $(1/m + O(\rho))n$, the attachment of v has incident edit distance at least cn from the clone set after decreasing ρ. Here the relevant attachment to the large classes is $A = [m]$, while the clone attachments are $[m] \setminus \{i\}$, $i \in [m]$. By Lemma A.1, after decreasing ρ, replacing v by the best clone among the m large classes gains at least $c_0 n^{k-1}$ induced copies containing v and otherwise using only the large classes.

It remains to check that the sequential moving does not lose this gain. At the step when j residual vertices have already been moved, the current large-part model differs from the original one in $O(jn)$ adjacencies. The rooted comparison for the next vertex can change only on k-sets containing that vertex and one of these changed adjacencies, so the same counting as in Lemma 3.1 gives an error $O_F(jn^{k-2})$. Since $j \le |R| \le \rho n$, this is $O_F(\rho n^{k-1})$. Copies containing v and another still-unmoved residual vertex contribute another $O_F(|R| n^{k-2}) = O_F(\rho n^{k-1})$. Choosing ρ small enough, both errors are absorbed by the $c_0 n^{k-1}$ rooted gain. Hence each moved vertex gives gain at least $c_1 n^{k-1}$.

Summing over all vertices of R gives a complete m-partite graph B with $N(F,B) - N(F,C) \geq c_2|R|n^{k-1}$. The edit distance from C to B is at most $2|R|n$, since each moved vertex changes at most n adjacencies and the edits among residual vertices are $O(|R|^2) \leq |R|n$. Therefore $N(F,B) - N(F,C) \geq c\,\mathrm{edit}(C,B)n^{k-2}$. The bound $|b_i - n/m| \leq \zeta n$ follows from $\big||V_i| - n/m\big| \leq \zeta n/4$ and $|R| \leq \zeta n/4$. □

The final bookkeeping proposition turns this cleanup into the finite reduction used earlier in Lemma 5.2.

Proposition A.3. *Let $F \coloneqq T(k,r)$ with $2 \leq r < k$, and let $m \coloneqq m(F)$. For every $\zeta > 0$ there are constants C_{clean}, $\varepsilon_{\text{clean}}$, and n_{clean}, depending only on F and ζ, such that the following holds for all $n \geq n_{\text{clean}}$. Let C be a complete multipartite n-vertex graph. Suppose $N(F,C) \geq I(F,n) - Dn^{k-2}$, where $0 \leq D \leq \varepsilon_{\text{clean}}n$. Then there is a complete m-partite graph $B \coloneqq K_{b_1,\dots,b_m}$ such that $\mathrm{edit}(C,B) \leq C_{\text{clean}}D$ and $|b_i - n/m| \leq \zeta n$ for every i.*

Proof. The limiting uniqueness theorem of Liu, Ma, and Zhu [12, Theorem 3.1], together with Lemma 6.1, puts $x(C)$ in the neighborhood of $x_\star$ required by Lemma A.2. Indeed, $Dn^{k-2} \leq \varepsilon_{\text{clean}}n^{k-1} = o(n^k)$, and hence this localization is uniform once n_{clean} is sufficiently large. Apply Lemma A.2 to obtain a complete m-partite graph $B = K_{b_1,\dots,b_m}$ with $|b_i - n/m| \leq \zeta n$ and $N(F,B) - N(F,C) \geq c\,\mathrm{edit}(C,B)n^{k-2}$. Since B is an n-vertex graph, $N(F,B) \leq I(F,n)$. Hence $c\,\mathrm{edit}(C,B)n^{k-2} \leq N(F,B) - N(F,C) \leq I(F,n) - N(F,C) \leq Dn^{k-2}$, and therefore $\mathrm{edit}(C,B) \leq c^{-1}D$. □

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